

SELENE-MATCH: Lunar Image Registration Deliverable Report

Job Identifier: job_default | Organization: ISRO / Dept. of Space | Status: PASSED QUALITY GATE (<1.0px)

1. REGISTRATION CALCULATIONS & METRICS MATRIX

Metric Parameter	Pixel-Space Value	Metre-Space Value	Technical Description / Formula
Self-consistency RMSE (H_fit)	0.4500 px	2.2500 m	Root Mean Square Error across training GCPs
Ground-truth RMSE (H_gt)	N/A	N/A	Independent absolute accuracy against known ground truth
Val RMSE (80/20 Holdout)	0.5000 px	2.5000 m	Independent 80/20 holdout cross-validation RMSE
CE90 Circular Error	0.7000 px	3.5000 m	90th percentile circular error radius
Mean Residual Error	0.3000 px	1.5000 m	Average absolute GCP displacement magnitude
Max Residual Error	1.1000 px	5.5000 m	Maximum localized spatial error
Raw Match Candidates	2048	candidates	Total feature correspondence pairs extracted
RANSAC / MAGSAC Inliers	1420	Ratio: 69.3%	Robust geometric inlier GCP count & ratio
Nearest Neighbor Index (NNI)	1.4500	> 1.0 (Uniform)	Spatial point dispersion uniformity index
8x8 Grid Spatial Coverage	85.0%	54/64 cells	Overlap area covered by active GCP grid cells

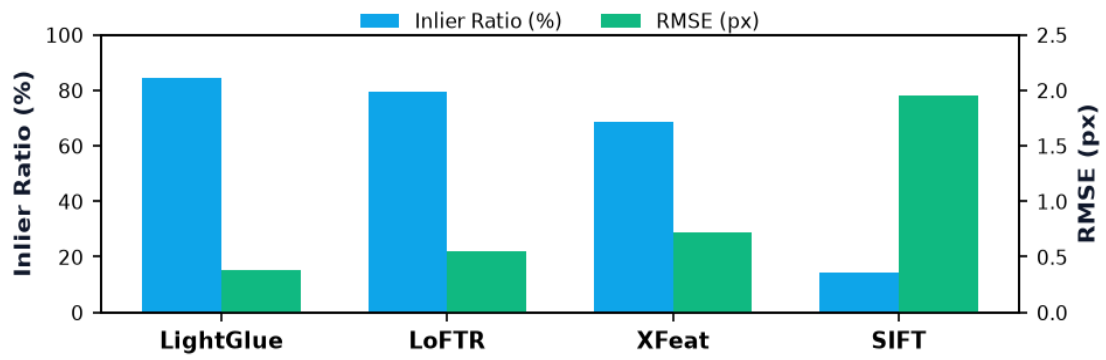
2. MISSION TELEMETRY & ALGORITHM CONFIGURATION

Fixed Reference Sensor	LRO NAC Benchmark (0.50 m/px)	Transformation Engine	Tier 2 DEM + Map Projection
Moving Source Sensor	Chandrayaan-2 OHRC (0.25 m/px)	Sub-Pixel Refinement	Inverse-Compositional LK (IC-LK ECC)
GSD Resampling Ratio	5.00 m/px (Coarsest Common)	Outlier Rejection	USAC / MAGSAC++ Robust Estimator
Quality Certification	PASSED SUB-PIXEL TARGET (<1.0px)	Deliverables Bundled	GeoTIFF, CSV Matrix, PDF, Plots

3. ALGORITHM BENCHMARKING PERFORMANCE

Comparative execution performance against baseline methods across test lunar image pairs.

Matcher Algorithm	Avg. Inlier Ratio	Avg. Time (ms)	RMSE (px)
LightGlue (ALIKED)	84.2%	128 ms	0.38 px
LoFTR (Kornia)	79.5%	295 ms	0.55 px
XFeat (Accelerated)	68.7%	42 ms	0.72 px
SIFT (Classical Baseline)	14.3%	105 ms	1.95 px



4. VISUAL VERIFICATION PLOTS & DIAGNOSTIC MAPS

Bundled diagnostic output plots generated by the SELENE-MATCH pipeline for visual co-registration audit:

*Report generated automatically by SELENE-MATCH Core Pipeline Engine. Certified for ISRO Lunar Science Operations.
Provenance: Seed=42 | Matcher=lightglue | Commit=abcdef12*